THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

033/2C BIOLOGY 2C

(Actual Practical)

TIME: 2:30 HOURS YEAR: 2025

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **all** questions.
3. Each question carries **twenty-five (25)** marks
4. Non programming calculators may be used.
5. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer sheets provided.

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1. You are provided with four test tubes labeled 1, 2, 3 and 4 a beaker, sieve measuring cylinder, test tube rack, specimen **P** and table reagents. Use them to carry experiment instructed in item (i) –(Ix), then answer questions that follow
   (i) Take specimen **P**, peel it to remove the outer cover and cut it into four pieces
   (ii) Grind two pieces of the specimen **P** by using mortar and pestle to obtain the paste
   (iii) Put the paste in the beaker and add 30mls of water and steer the mixture
   (iv) Filter by using a sieve to obtain solution P
   (v) Put 2ml of solution **P** into each test tube 1,2,3, and 4
   (vi) Add 2drops of iodine solution into test tube 1
   (vii) Add 2ml of Benedict's solution into test tube 2 and boil the content
   (viii) Add 1ml of dilute hydrochloric acid into the test tube 3 and boil the mixture and allow the mixture to cool. After cooling add 2ml of sodium hydroxide solution slowly until the fizzing stops followed by 2ml of Benedict's solution and boil the mixture again.
   (ix) Add 1ml of sodium hydroxide solution into test tube 4, followed by 2-3 drops copper(ii)sulphate solution

QUESTIONS

a) What is the aim of experiment?
b) Based on observations made in test tube 1, 2, 3 and 4. What are the types of food substances contained in the specimen **P**. Give reasons to justify your answers.
c) Why sodium hydroxide and dilute hydrochloric acid was used in this experiment
d) Why boiling and cooling is important in procedure (viii)
e) What will be the end product of digestion of food substances contained in specimen **P** in human body
f) What is the importance of the food substances identified in specimen **P** in human body
g) Why the skills used in this experiment useful for preparation of balanced diet in your daily life.

2. You have been provided with specimens **A, B** and **C**

a) i) Identify the specimens **A, B** and **C** by their common names.

ii) Classify specimen **B** and **C** to phylum level

iii) Name the class and state two distinctive features for each class of specimens **A, B** and **C**

b) i) State the habitat of specimen **A** and **C**

ii) State the adaptation features of specimen **B** to its mode of life

iii) In what ways are specimen **B** and **C** useful and harmful to man and his/her environment.